Particles World:Thermal Mix & Evaporation - Educational Models in NetLogo

Introduction

This project showcases a set of interactive models developed in NetLogo (Wilensky, 1999), aimed at enhancing the understanding of particle interactions in middle school science classes. It demonstrates the application of computational models to simplify complex scientific concepts.

Audience and Setting

Designed for middle school students, these models serve as an introductory tool in science classes, providing a hands-on learning experience. They offer an engaging approach to explore fundamental scientific principles in a classroom setting.

Project Goals and Learning Objectives

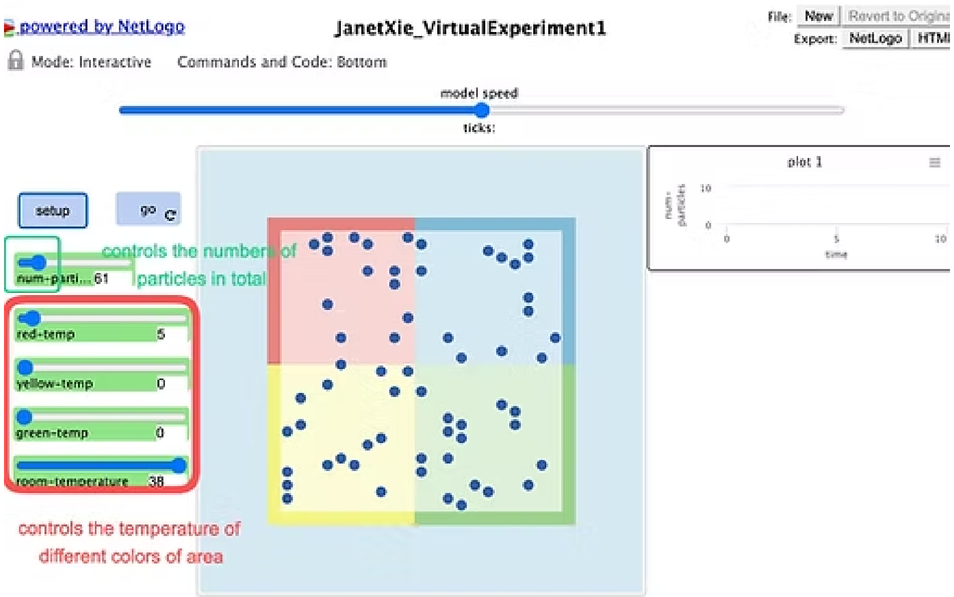
The primary goal of this project was to facilitate a deeper understanding of particle behavior under varying temperatures, states of matter, and the impact of environmental changes on these particles.

The learning objectives focused on enabling students to visually comprehend and analyze these concepts through interactive models.

Project Description

The project comprises two NetLogo models based on the Molecular Dynamics Lennard-Jones model (Kelter & Wilensky, 2022). The 'Thermal Mix' model simulates particle interactions at different temperatures, while the 'Evaporation' model demonstrates the process and effects of evaporation. Both models are designed to make abstract scientific concepts more tangible and understandable for students."

Software Specification and Applications



Model 1: Thermal Mix

The model aims to visually represent how particles interact and behave under different temperature conditions. It serves as an educational tool to help students understand concepts like thermal energy, particle motion, and temperature effects on matter.

WHAT IS IT?

This model is designed to show how particles behave in different temperature environments. The model consists of a container divided into four parts: red, yellow, green, and blue. The temperature of the red, yellow, and green parts can be adjusted using the sliders, while the temperature of the blue parts is determined by the room temperature. The model allows you to explore how changing the temperature affects the behavior of the particles.

HOW IT WORKS

The movement of the particles in this model is based on the Lennard-Jones potential. This means that the particles follow specific rules to interact with each other based on their distance and energy levels. The particles will move and interact with each other based on their temperature and position within the container.

HOW TO USE IT

To use this model,you can adjust the number of particles in the box and use temperature sliders for the red, yellow, and green parts of the container. They can also adjust the room temperature slider to see how it affects the behavior of all particles. Once the temperatures have been set, you can click the "setup" button to initialize the model, and then click the "go" button to start the simulation.

THINGS TO NOTICE

What happens when only one parts of the container is set to the minimal/maximum temperature? Why is that?

THINGS TO TRY

Notice how the particles move and interact with each other based on the temperature of the different parts of the container. They can also observe how the room temperature affects the behavior of the particles in the container. What happens when What happens when there's only one particle? You can play around with different numbers of particles.

EXTENDING THE MODEL

To extend this model, you can add more particles or adjust the Lennard-Jones potential to make the model more detailed and accurate. They can also change the size or shape of the container to create different environments for the particles.

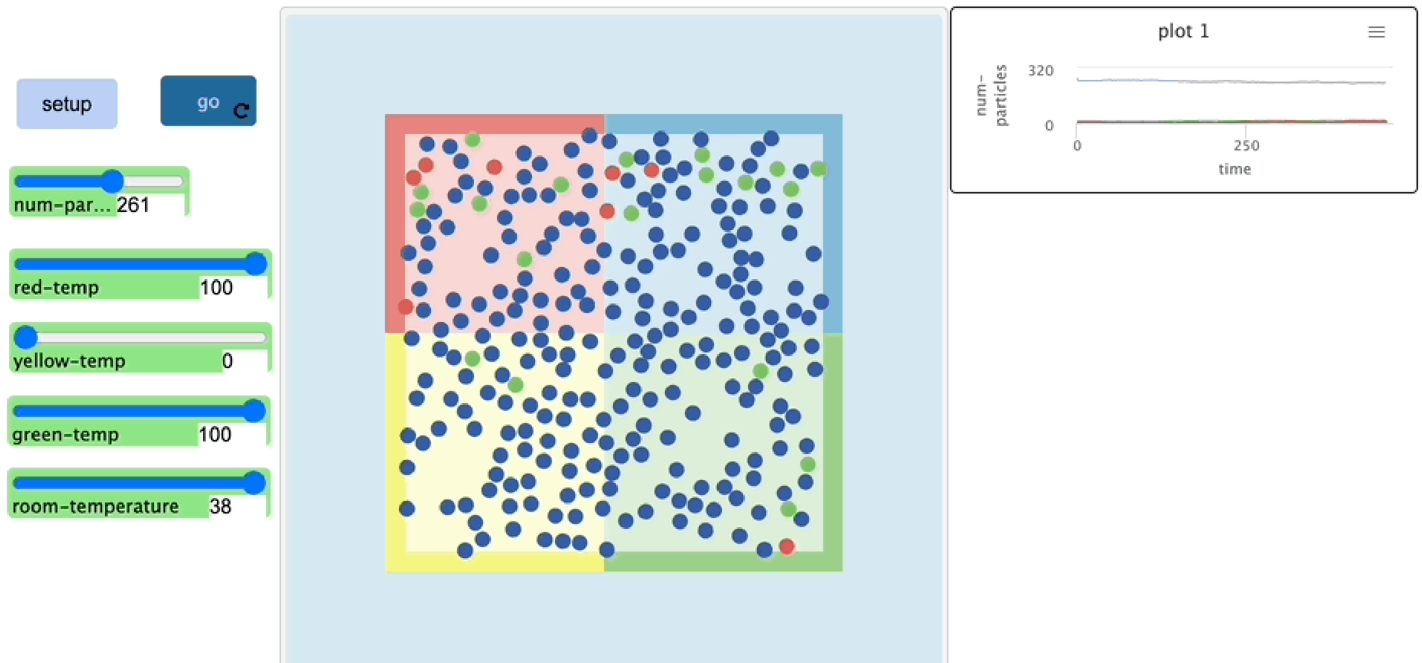
Example Code

```
to move-particles  ; particles procedure
  ;; Uses velocity-verlet algorithm
  let new-xcor velocity-verlet-pos xcor vx fx
  let new-ycor velocity-verlet-pos ycor vy fy
  if new-ycor > 16  [die] ;;if particles have an intention to move outside the wall, it will disappear

  ifelse [wall?] of patch new-xcor ycor [ ;;if particles hit the wall
    let x-direction ifelse-value vx > 0 [-1][1] ;;the velocity will be changed into negative number in order to bounce back
    set vx x-direction * new-velocity (patch new-xcor ycor)
  ][

    set xcor new-xcor
  ]
  ifelse [wall?] of patch xcor new-ycor [
    let y-direction ifelse-value vy > 0 [-1][1]
    set vy y-direction * new-velocity (patch xcor new-ycor)

  ][set ycor new-ycor
  ]

  color-particles
end


to color-particles ;;to use different colors to symbolize the current temperarture of a particle
  let speed sqrt (vx ^ 2 + vy ^ 2)
  if speed > 1.5 [set color 16] ;;particles with a higher temperature (higher speed) will turn into red
  if speed < 1.5 [set color 56] ;;particles with a medium temperature (medium speed) will turn into green
  if speed < 1 [set color blue] ;;particles with a room temperature (regular speed) will turn into blue as initilized
end
```

Example

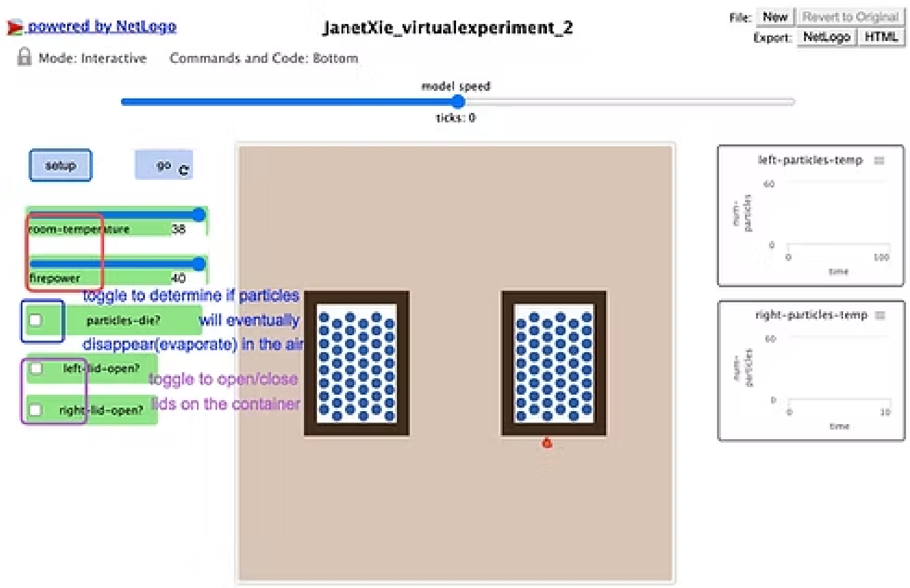
Trajectory

As the temperature sliders of different areas are adjusted by the user, the particles move differently according to the temperature whereas more red particles(high-temperature particles) appear within the higher temperature area and more blue particles(temperature particles appear with the lower temperature area.)


setup
go
num-par... 261
red-temp 100
yellow-temp 0
green-temp 100
room-temperature 38
plot 1
320
0
num-particles
0
250
time

Model 2: Evaporation

This model vividly demonstrates how water molecules transition from liquid to gas under varying temperature conditions. Key features include adjustable temperature and firepower sliders, and switches for opening or closing container lids. These elements allow students to experiment with and observe the rate and mechanics of evaporation in real-time, providing a deeper understanding of this essential physical process.



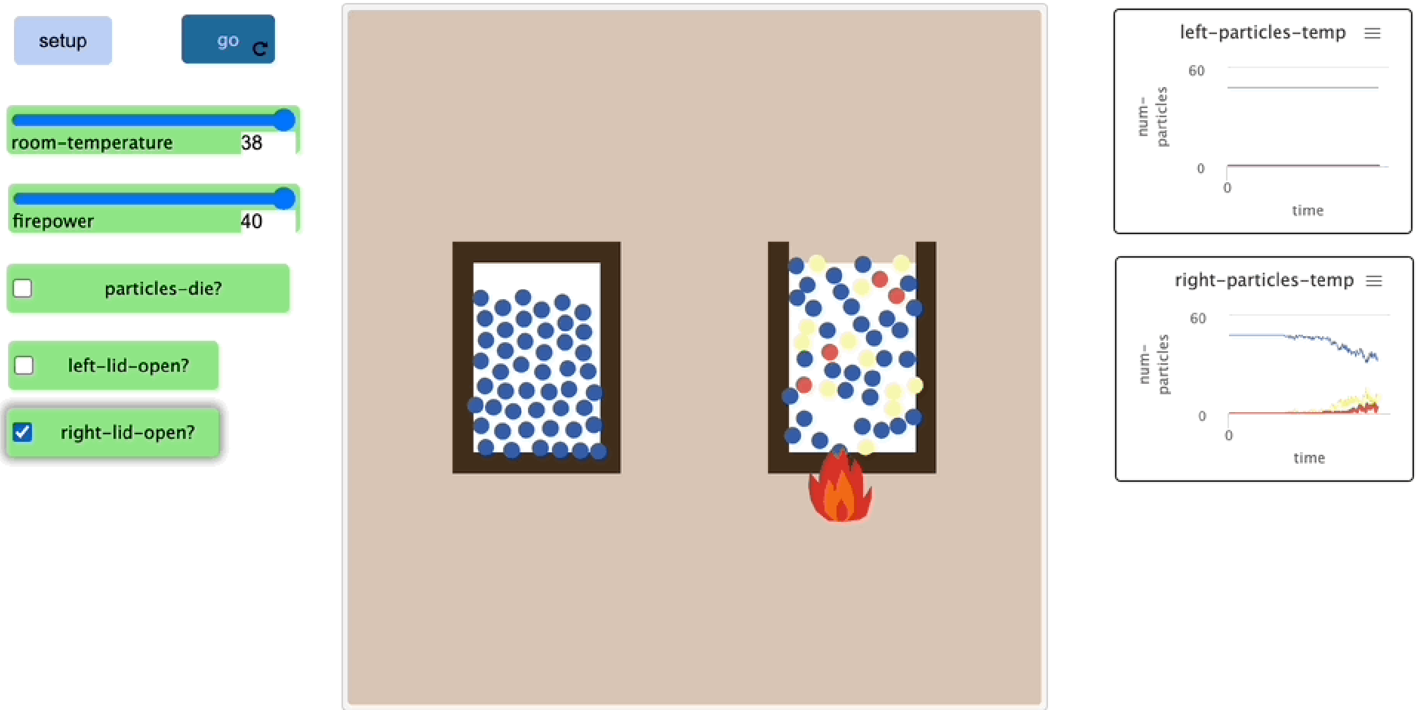
Example Code

```
to setup-firepower ;; create fire shape under the right container
  create-fires 1 [
    set shape "fire"
    set color red
    setxy 40 -39
  ]

end

to lid-control ;;to open/close lids on the container so the particles could escape into the air/stay inside the container
  ifelse left-lid-open? [ask patches with [
    (pxcor >= -10 and pxcor <= -5) and (pycor = 5)]
    [set pcolor 38 set wall? false ]]
  [ask patches with [
    (pxcor >= -10 and pxcor <= -5) and (pycor = 5)]
    [set pcolor 32 set wall? true ]]
  ifelse right-lid-open? [ask patches with [
    (pxcor >= 5 and pxcor <= 10) and (pycor = 5)]
    [set pcolor 38 set firewall? false set wall? false]]
  [ask patches with [
    (pxcor >= 5 and pxcor <= 10) and (pycor = 5)]
    [set pcolor 32 set firewall? true set wall? true]]
end

to turn-on-fire ;;to set a variable as a additional temperature factor in order to obeserve faster evaporation
  ifelse firepower > 0 [ifelse size = (firepower / 10) [set size firepower / 10 + 0.5] [set size firepower  / 10]]
  [set size 0]
end
```

Example

Trajectory

As the firepower is set to the highest on the right container, the particles quickly evaporate in the air while the particles in the closed-lid left container remain regular speed. When the lid on the right container is closed, the particles move into the air keep moving and are influenced by the room-temperature. If the particles-die? toggle is selected, the particles in the air will remain moving and might fall into the left container and create temperature change in the left container.



Student Engagement and Learning Trajectories

The models are structured to encourage exploration, allowing students to manipulate variables and observe outcomes. This interactive approach facilitates a progressive understanding, from basic concept recognition to advanced analysis and application, thereby supporting various stages of learning.

Reference

Kelter, J. and Wilensky, U. (2022). NetLogo Molecular Dynamics Lennard-Jones model. http://ccl.northwestern.edu/netlogo/models/MolecularDynamicsLennard-Jones. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Wilensky, U. (1999). NetLogo. http://ccl.northwestern.edu/netlogo/. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.